

Proofs for “VioExplain: Single-Event Knowledge Explains Concurrent Industrial Constraint Violations”

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This document states and proves the theoretical results for VioExplain. Theorem numbers correspond to the main text. Each statement includes the assumptions used in its proof. Theorem 1 concerns an event-prior-penalized profile likelihood, Theorem 2 an error lower bound for rules that read a window through a fixed interface, Theorem 3 a recovery certificate, and Theorem 4 a staged calibration protocol. Lemmas and remarks of this document are numbered S1, S2 and so on.

1 Notation and the Path of Algorithm 1

A window X has a description $\phi(X) \in \mathbb{R}^D$ and a set $\mathcal{V}(X)$ of violation features. A feature records a constraint key together with its numerical degree interval. The candidate events form a finite, nonempty set $\mathcal{E}$, with $|\mathcal{E}| = K$. For an event E , $R(E) = R^*(E) \cup R^+(E)$ is its collection of exact and possible representations; in the optimization model, $R(E)$ also denotes their incident units. The support $\text{Supp}(E)$ and footprint $f_E(u)$ used by the algorithm are fixed learned quantities. The temporal posterior is denoted by $p^t(X)$ and the description-based posterior by $p^s(X)$. The fused posterior is $p(X) \propto (p^s(X))^{1-\lambda}(p^t(X))^\lambda$, normalized over normal operation and the candidate events.

Under a shared noise stream ξ , $X^{(H)}(\xi)$ denotes the run with active events H , and $X^{(0)}(\xi)$ the fault-free run. A violation feature in $\mathcal{V}(X^{(H)}) \setminus \mathcal{V}(X^{(0)})$ is caused. An event $E \in H$ is effective if at least one caused feature disappears when E is removed, or is present in the single-event run of E . All comparisons use the same specified violation interface and paired window position.

Put $S_0 = \emptyset$ and $r_0 = \phi(X)$. If $p_\emptyset(X) \geq \tau_0$, the event output is empty. Otherwise select $E_1 \in \arg \max_{E \in \mathcal{E}} p_E(X)$, set $S_1 = \{E_1\}$, and put $r_1 = r_0 - \hat{\Delta}_{E_1}(r_0)$. At a later step j , stop if $|S_j| = K_{\max}$ or no unselected candidate remains. Otherwise select

$$E_{j+1} \in \arg \max_{E \in \mathcal{E} \setminus S_j} z_E(X, r_j, S_j).$$

If its score is at most τ_1 , stop and return S_j . If its score is greater than τ_1 , add it and deduct its footprint, giving $S_{j+1} = S_j \cup \{E_{j+1}\}$ and $r_{j+1} = r_j - \hat{\Delta}_{E_{j+1}}(r_j)$. Here $1 \leq K_{\max} \leq K$, and ties use a fixed measurable rule. A violation on unit u is assigned to an event maximizing $f_E(u)$ among $E \in S$ with $u \in \text{Supp}(E)$, or to background if there is no such event. The unknown flag is set when the template deviation $D_S(\phi(X))$ exceeds τ_u . The deviation, for example a Mahalanobis distance using fixed template means and a fixed positive-definite covariance, is assumed finite and measurable.

2 Theorem 1

2.1 Model and scope

Let U and $\mathcal{E}$ be finite sets of units and candidate events. For each event, $R^*(E) \subseteq R(E) \subseteq U$ specifies its exact units and all its admissible units. The following assumptions define a family of conditional observation models and an event prior.

- (M1) Events have independent prior probabilities $0 < \pi_E \leq \frac{1}{2}$. Write $w_E = \log((1 - \pi_E)/\pi_E) \in [0, \infty)$.
- (M2) For $H \subseteq \mathcal{E}$, the admissible assignments form $\mathcal{A}(H)$. An assignment $a : U \rightarrow H \cup \{\perp\}$ may assign u to E only if $u \in R(E)$. Background assignment is permitted except on $R^*(H) = \bigcup_{E \in H} R^*(E)$. Thus an exact unit must be assigned to a selected event, but need not be assigned to its exact owner. The assignment is a nuisance parameter over which we maximize; no probability distribution on assignments is assumed.
- (M3) At the observed vector $x = (x_u)_{u \in U}$, the conditional likelihood is $p(x | H, a) = \prod_u p_u(x_u | a(u))$, with background density or mass function $p_u^0 = p_u(\cdot | \perp)$. The relevant density or mass values at x are positive and finite, and all $g_u(E) = \log(p_u(x_u | E)/p_u^0(x_u))$, for $u \in R(E)$, are finite real numbers. Put $g_u(\perp) = 0$.

For a possible representation with response probability $q_{Eu} \in (0, 1]$, the mixture $p_u(x | E) = q_{Eu}p_u^{\text{resp}}(x) + (1 - q_{Eu})p_u^0(x)$ gives $g_u(E) \geq \log q_{Eu} + \log(p_u^{\text{resp}}(x_u)/p_u^0(x_u))$ whenever the latter is finite. The results below use the actual $g_u(E)$, rather than replacing it by this lower bound.

Define $m_u(H) = \max\{g_u(E) : E \in H, u \in R(E)\}$, with $\max \emptyset = -\infty$, and

$$v_u(H) = \begin{cases} m_u(H), & u \in R^*(H), \\ \max(0, m_u(H)), & u \notin R^*(H). \end{cases}$$

The event-prior-penalized profile log-likelihood ratio is

$$F_{\text{ex}}(H) = \log \frac{\Pr(H) \max_{a \in \mathcal{A}(H)} p(x | H, a)}{\Pr(\emptyset) \prod_u p_u^0(x_u)}.$$

This score is not a joint or marginal posterior ratio for a probabilistic latent assignment. Such a posterior would require an assignment distribution and its normalization. Define its separable surrogate by

$$F(H) = \sum_{u \in U} \max(0, m_u(H)) - \sum_{E \in H} \hat{w}_E, \quad \hat{w}_E = w_E + \sum_{u \in R^*(E)} [-g_u(E)]_+.$$

Here $[t]_+ = \max(t, 0)$.

Theorem 1 (Event-prior-penalized profile likelihood). *Under (M1)–(M3), $F_{\text{ex}}(H) = \sum_u v_u(H) - \sum_{E \in H} w_E$ and $F(H) \leq F_{\text{ex}}(H)$ for every H . Equality holds if and only if, for every $u \in R^*(H)$, one of the following conditions holds: (i) $m_u(H) \geq 0$ and every exact owner of u in H has nonnegative gain; (ii) $m_u(H) < 0$, exactly one event in H is an exact owner of u , and its gain equals $m_u(H)$.*

The coverage term in F is monotone submodular and its penalty is modular. Put

$$G_u = \max_{E: u \in R(E)} [g_u(E)]_+,$$

with $G_u = 0$ if no event is incident to u . Then $C(H) = \sum_u G_u - F(H)$ is the minimum assignment cost plus opening cost of a nonmetric uncapacitated facility location instance with opened event set H , event opening costs $\hat{w}_E$, admissible assignment costs $c(u, E) = G_u - [g_u(E)]_+$, and an always-open background facility with $c(u, \perp) = G_u$ and zero opening cost. When finite rational gains and costs are the optimization input, maximizing F is NP-hard. For every fixed $\epsilon \in (0, 1)$, a polynomial-time $(1 - \epsilon) \ln |U|$ approximation of its minimum facility cost on all sufficiently large instances would imply $\text{P} = \text{NP}$.

Proof. Step 1, the profile score. Independence of the event priors gives $\Pr(H)/\Pr(\emptyset) = \exp(-\sum_{E \in H} w_E)$. For an admissible assignment the log conditional likelihood ratio is $\sum_u g_u(a(u))$. Assignments on different units impose no joint constraint. Therefore maximizing over $a \in \mathcal{A}(H)$ chooses the best permitted source separately on each unit, giving $v_u(H)$ and the stated expression for F_{ex} . No assignment prior is included in this calculation, and no posterior interpretation is asserted.

Step 2, the lower bound and exact equality conditions. Fix H . If $u \notin R^*(H)$, both scores have the same unit contribution and there is no exact-owner penalty. For $u \in R^*(H)$, write $A_u = \{E \in H : u \in R^*(E)\}$,

$m = m_u(H)$ and $P_u = \sum_{E \in A_u} [-g_u(E)]_+$. The difference between the profile contribution and the surrogate contribution is

$$d_u = m - \max(0, m) + P_u.$$

If $m \geq 0$, then $d_u = P_u \geq 0$, with equality exactly when every exact owner has nonnegative gain. If $m < 0$, all covering-event gains are negative. Set $t = -m > 0$. Every exact owner satisfies $-g_u(E) \geq t$, so $P_u \geq |A_u|t \geq t$ and $d_u = P_u - t \geq 0$. Equality requires and is implied by $|A_u| = 1$ and the unique exact owner's gain being m . Other selected events may cover the unit with gains at most m . Since all d_u are nonnegative, their sum is zero exactly under the stated unitwise conditions.

Step 3, submodularity. Define $a_{uE} = [g_u(E)]_+$ for $u \in R(E)$ and $a_{uE} = 0$ otherwise. The coverage term is $\sum_u \max_{E \in H} a_{uE}$, with empty maximum zero. For $A \subseteq B$ and $E \notin B$ its unitwise marginal gains satisfy

$$\left[a_{uE} - \max_{E' \in A} a_{uE'} \right]_+ \geq \left[a_{uE} - \max_{E' \in B} a_{uE'} \right]_+.$$

They are nonnegative, proving monotonicity and submodularity of coverage. Each $\hat{w}_E$ depends on E and the fixed observation, not on H , so the penalty is modular. The full function F need not be monotone.

Step 4, facility location. All opening and assignment costs are nonnegative. In this facility instance, background assignment is permitted for every unit, including exact units; the exact-unit penalties have already been incorporated into opening costs. For a fixed opened set H , the best assignment cost is

$$\sum_u \min(\{G_u\} \cup \{G_u - [g_u(E)]_+ : E \in H, u \in R(E)\}) = \sum_u G_u - \sum_u \max(0, m_u(H)).$$

Adding opening costs gives $C(H)$. These costs need not obey any metric condition.

Step 5, computational hardness. Take a set-cover instance with nonempty universe U , $n = |U|$, and m subsets $S_1, \dots, S_m$ whose union is U . Set $R^*(E_i) = R(E_i) = S_i$, $w_{E_i} = 1$, and $g_u(E_i) = G = mn + 1$ on each incidence. These are finite rational optimization inputs. They also have likelihood realizations: use background $\mathcal{N}(0, 1/(2G))$, event response $\mathcal{N}(1, 1/(2G))$, observation $x_u = 1$, and event prior $1/(1+e)$. Then

$$F(H) = G \left| \bigcup_{E_i \in H} S_i \right| - |H|, \quad C(H) = G \left| U \setminus \bigcup_{E_i \in H} S_i \right| + |H|.$$

A noncover costs at least $G > m$, whereas opening all events costs m , so minimum facility cost equals minimum cover cardinality. This proves NP-hardness. For fixed $\epsilon \in (0, 1)$ and sufficiently large n , put $\rho = (1 - \epsilon) \ln n < n$. A ρ -approximate solution costs at most $\rho m < G$, hence must cover U and gives a set cover within factor ρ . The set-cover inapproximability theorem therefore proves the stated claim [1]. $\square$

Remark S1 (Approximation of the facility cost). There is a polynomial-time density-greedy algorithm with approximation factor $H(\Delta) = \sum_{i=1}^{\Delta} 1/i$, where $\Delta = \max(\{1\} \cup \{|R(E)| : E \in \mathcal{E}\})$. This is an algorithm for the nonnegative facility cost, not a guarantee for Algorithm 1 of the main text.

To prove the claim, create a weighted set-cover column for every nonempty star (E, T) , $T \subseteq R(E)$, with covered set T and cost $\hat{w}_E + \sum_{u \in T} c(u, E)$. Also create a background singleton of cost G_u for each u . Every facility solution decomposes into one star for the units assigned to each used event and background singletons, at the same cost after unused opened events are removed. Conversely, a cover by stars gives a facility solution of no greater cost: open each event used by a star once, and assign each unit to one of its covering stars, retaining its indicated facility. The removed duplicate opening costs and unused assignment costs are nonnegative. Hence the two optimum costs coincide.

The weighted set-cover density-greedy bound is $H(\Delta)$ [2], since every column has at most Δ units. Although the number of stars is exponential, an exact greedy step is polynomial. For a fixed event, deleting already covered units from a proposed star cannot increase its cost per newly covered unit. Sort its remaining admissible units by $c(u, E)$; among stars with exactly k newly covered units, a prefix of length k minimizes cost. Enumerating all nonempty prefixes for every event and all remaining background singletons finds the global minimum-density star [3]. Each iteration covers a new unit, so there are at most $|U|$ iterations. The argument includes zero-cost stars. If U is empty, the empty event set is optimal with cost zero.

A cost approximation factor does not imply a multiplicative guarantee on F . Write $B = \sum_u G_u$, $C^* = \min_H C(H)$ and $F^* = \max_H F(H) = B - C^*$. If $C(H) \leq \rho C^*$, then only the additive implication $F(H) \geq F^* - (\rho - 1)C^*$ follows in general.

3 Theorem 2

Theorem 2 (Interface lower bound on paired runs). *Let P and Q be window laws, let T be a measurable interface, and let f be a measurable decision rule that uses the window only through $T(X)$. A randomized rule may additionally use a random variable independent of the window whose law is the same under both hypotheses. For every measurable decision event $\mathcal{A}$,*

$$P(f \notin \mathcal{A}) + Q(f \in \mathcal{A}) \geq 1 - \text{TV}(T_{\#}P, T_{\#}Q).$$

For every coupling (X, X') with marginals P and Q ,

$$\text{TV}(T_{\#}P, T_{\#}Q) \leq \Pr(T(X) \neq T(X')).$$

In particular, paired runs $X^{(h)}(\xi)$ and $X^{(h')}(\xi)$ with the specified marginal window laws give a lower bound of $1 - \delta_T(h, h')$ on the sum of the two errors, where

$$\delta_T(h, h') = \Pr_{\xi}(T(X^{(h)}(\xi)) \neq T(X^{(h')}(\xi))).$$

This is a population probability over the shared noise ξ .

Proof. Put $\mu = T_{\#}P$ and $\nu = T_{\#}Q$. If the rule uses independent randomization W with law λ , the event $\{f \in \mathcal{A}\}$ is a measurable set A in the product of the interface and randomization spaces. Each section A_w satisfies $|\mu(A_w) - \nu(A_w)| \leq \text{TV}(\mu, \nu)$. Consequently,

$$P(f \in \mathcal{A}) - Q(f \in \mathcal{A}) = \int [\mu(A_w) - \nu(A_w)] \lambda(dw) \leq \text{TV}(\mu, \nu).$$

Rearranging gives the two-point bound [4]; a deterministic rule is the special case of a constant W .

For any measurable set B in the interface space and any coupling,

$$\begin{aligned} \mu(B) - \nu(B) &= \Pr(T(X) \in B) - \Pr(T(X') \in B) \\ &\leq \Pr(T(X) \in B, T(X') \notin B) \\ &\leq \Pr(T(X) \neq T(X')). \end{aligned}$$

Taking the supremum over B proves the coupling inequality. Applying it to the paired runs proves the last assertion. Under equal priors, minimization over all binary decisions on the interface gives the Bayes error $\frac{1}{2}(1 - \text{TV}(\mu, \nu))$, by the variational characterization of total variation. $\square$

Remark S2 (Estimating the paired probability). The theorem requires no parametric family for the marginal laws. A mismatch fraction from finitely many paired runs estimates δ_T ; it is not the population probability itself. A finite-sample confidence statement about the error lower bound must use a valid upper confidence bound for δ_T , with the sampling units and dependence of the paired observations accounted for.

Remark S3 (Two interfaces referred to in the main text). Interval membership. Let P_{E_1} and P_{E_2} denote the laws of a continuous description vector ϕ under two events. Fix common intervals I_u for its coordinates, let $B = \prod_u I_u$, and define the common membership interface $T_I(\phi) = (\mathbf{1}[\phi_u \in I_u])_u$. If $P_{E_i}(B) \geq 1 - \epsilon$ for $i = 1, 2$, with $0 \leq \epsilon \leq 1$, every decision rule based only on T_I has error at least $\frac{1}{2}(1 - \epsilon)$ under equal priors, whereas the Bayes error with access to the full description is $\frac{1}{2}(1 - \text{TV}(P_{E_1}, P_{E_2}))$. Indeed, the interface has a common value t_0 throughout B , so its laws μ_1 and μ_2 each put mass at least $1 - \epsilon$ on t_0 . If a measurable set A contains t_0 , both $\mu_i(A)$ lie in $[1 - \epsilon, 1]$, and otherwise both lie in $[0, \epsilon]$. Thus $\text{TV}(\mu_1, \mu_2) \leq \epsilon$, and Theorem 2 gives the error bound. The variational characterization applied to the description laws

gives the full-description Bayes error. The statement concerns the specified common membership interface. Membership in additional intervals need not be constant on B , so it does not extend to such a richer interface without a separate argument.

Violated constraints. Let $T_{\text{key}}(X) = \{\text{key}(v) : v \in \mathcal{V}(X)\}$ be the set of violated constraints of a window, which discards the numerical degree intervals. Apply Theorem 2 with $T = T_{\text{key}}$, the window laws of an event E and of the fault-free run, and the decision event $\mathcal{A} = \{F : E \in F\}$. Every window-level rule f that depends only on T_{key} satisfies

$$\Pr_E(E \notin f) + \Pr_{\emptyset}(E \in f) \geq 1 - \delta_{\text{key}}(E, \emptyset),$$

where $\delta_{\text{key}}(E, \emptyset)$ is the population probability that the paired event and fault-free windows differ in their sets of violated constraints. The same argument with $T = \mathcal{V}(\cdot)$ gives the bound $1 - \delta_{\mathcal{V}}(E, \emptyset)$ for rules that read the complete violation features, where equality of feature sets includes equality of the recorded degree intervals. Two windows can have the same violated constraints and different degree intervals, so $\delta_{\text{key}} \leq \delta_{\mathcal{V}}$ and the inequality may be strict. The bound for T_{key} therefore does not apply to a rule that also reads interval values. An application to a particular method must establish its interface and use the mismatch probability for that interface.

Remark S4 (Several windows of one run). For n independent and identically distributed interface observations under each hypothesis, let their one-window laws be μ, ν and let $A(\mu, \nu) = \int \sqrt{pq}$ be their Hellinger affinity relative to a common dominating measure. The inequalities $1 - \text{TV}(\mu, \nu) \geq \frac{1}{2}A(\mu, \nu)^2$ and $A(\mu^{\otimes n}, \nu^{\otimes n}) = A(\mu, \nu)^n$ give a lower bound $\frac{1}{2}A(\mu, \nu)^{2n}$ on the sum of the two errors. Consecutive windows need not be independent, so this product bound does not apply to them without further assumptions.

4 Theorem 3

Fix a window $X = X^{(H)}(\xi)$ with a nonempty true set $H \subseteq \mathcal{E}$ and $1 \leq k = |H| \leq K_{\text{max}}$. The candidate set $\mathcal{E}$ is finite, and the thresholds τ_0, τ_1 in this section are finite. In addition to the stopping rules in Algorithm 1, the algorithm stops if no unselected candidate remains. For a selected set $S \subseteq H$, define the paired reference description

$$\psi_S = \phi(X^{(H \setminus S)}(\xi)).$$

Thus $\psi_{\emptyset} = \phi(X)$ and $\psi_H = \phi(X^{(0)}(\xi))$. Write $z_E^{X,S}(r)$ for the addition score with the original window X and the selected set S held fixed. Only its residual argument r changes in the following comparison. At a visited prefix S_j with residual r_j , let

$$\omega_{S_j} = \max_{E \in \mathcal{E} \setminus S_j} |z_E^{X, S_j}(r_j) - z_E^{X, S_j}(\psi_{S_j})|,$$

with $\omega_{S_j} = 0$ if the candidate set is empty. All posterior and addition scores are finite. For maxima of scores over an empty set, use $\max \emptyset = -\infty$. Define

$$\eta_{\emptyset} = \max_{E \in H} p_E(X) - \max_{E' \in \mathcal{E} \setminus H} p_{E'}(X).$$

For $\emptyset \neq S \subsetneq H$ and $E \in H \setminus S$, put

$$\begin{aligned} b_E(S) &= z_E^{X,S}(\psi_S) - \max_{E' \in \mathcal{E} \setminus H} z_{E'}^{X,S}(\psi_S), \\ \ell_E(S) &= \min\{b_E(S), z_E^{X,S}(\psi_S) - \tau_1\}, \\ \eta_S &= \max_{E \in H \setminus S} \ell_E(S). \end{aligned}$$

The comparison with false candidates is then automatically satisfied when $H = \mathcal{E}$. When $\mathcal{E} \setminus H$ is nonempty, define the stopping margin

$$\eta_H = \tau_1 - \max_{E \in \mathcal{E} \setminus H} z_E^{X,H}(\psi_H).$$

Theorem 3 (Recovery certificate). *Suppose $p_\emptyset(X) < \tau_0$, $\eta_\emptyset > 0$, and $\eta_S > 2\omega_S$ for every nonempty proper prefix $S \subsetneq H$ visited by AEC-X. If $k < K_{\max}$ and $\mathcal{E} \setminus H$ is nonempty, also suppose that $\eta_H > \omega_H$ when the algorithm reaches H . Then AEC-X returns exactly H . Every violation on a unit that belongs to the support of exactly one selected event $E \in H$ is assigned to E .*

Proof. The condition $p_\emptyset(X) < \tau_0$ passes the initial background test. A true event attains $\max_{E \in H} p_E(X)$, and $\eta_\emptyset > 0$ makes its posterior larger than that of every false event. If no false event exists, every candidate is true. In either case, the first selection belongs to H and $S_1 \subseteq H$.

Suppose the algorithm has selected $S_j \subsetneq H$ with $1 \leq j < k$. Let $E^\dagger \in H \setminus S_j$ attain η_{S_j} . When a false event E' exists, the definitions give

$$z_{E^\dagger}^{X, S_j}(r_j) - z_{E'}^{X, S_j}(r_j) \geq \eta_{S_j} - 2\omega_{S_j} > 0.$$

Irrespective of whether false events exist, they also give

$$z_{E^\dagger}^{X, S_j}(r_j) \geq \tau_1 + \eta_{S_j} - \omega_{S_j} > \tau_1.$$

The highest-scoring unselected candidate therefore belongs to $H \setminus S_j$ and passes the addition threshold. The candidate set is nonempty because $H \setminus S_j$ is nonempty, and the cardinality bound does not stop the algorithm because $j < k \leq K_{\max}$. Thus $S_{j+1} \subseteq H$, proving the induction.

After k additions the selected set is H . If $k = K_{\max}$, the cardinality bound stops the algorithm. If $H = \mathcal{E}$, the absence of candidates stops it. In the remaining case, for every unselected candidate E ,

$$z_E^{X, H}(r_k) \leq z_E^{X, H}(\psi_H) + \omega_H \leq \tau_1 - \eta_H + \omega_H < \tau_1,$$

so the threshold stops it. Each stopping case returns exactly H . For a violation on a unit supported by exactly one event of H , the assignment rule has that event as its sole candidate and assigns the violation to it. $\square$

Remark S5 (Scope of the certificate). This is a sufficient condition along the path actually visited. It certifies the returned event set and the stated assignment behavior. Unique membership in a learned support does not itself establish that the assigned event is a counterfactual cause, nor does the theorem establish that every background violation is left unassigned.

An order-uniform condition is also possible. For each $S \subseteq H$, keep X , S , and the reference ψ_S fixed, and let $\bar{\omega}_S$ be the supremum of the same score discrepancy over residuals formed by successively deducting the events of S in every possible order. Conditions $\eta_S > 2\bar{\omega}_S$ for every nonempty proper subset of H , together with the initial conditions and the stopping condition $\eta_H > \bar{\omega}_H$ when needed, imply the pathwise certificate. Checking only the prefixes of one prescribed order does not establish this order-uniform condition. No global Lipschitz property of the learned scores is assumed; in particular, discontinuous tree scores need not satisfy such a property.

Remark S6 (Interaction of events). The certificate bounds the interaction of events through ω_S , and single-event information alone cannot replace such a bound. Use deterministic one-sample windows in $\mathbb{R}^3$, with fault-free window 0 and single-event windows e_1, e_2, e_3 for three events A, B, C , where e_i are the coordinate basis vectors and each coordinate has the normal interval $[-1/2, 1/2]$. One process sums the active single-event vectors. A second process agrees with it on every event set except $\{A, B\}$, for which the window is e_3 . The two processes have the same fault-free runs, single-event runs and sums of single-event effects. In the second process the windows of $\{A, B\}$ and of $\{C\}$ coincide while both A and B are effective, so no rule based on the window separates the two event sets.

5 Theorem 4

For $0 < \alpha < 1$ and real scores $z_1, \dots, z_n$, define the upper conformal threshold

$$Q_\alpha(z_1, \dots, z_n) = \begin{cases} z_{(m)}, & m = \lceil (1 - \alpha)(n + 1) \rceil \leq n, \\ +\infty, & m = n + 1, \end{cases}$$

where $z_{(m)}$ is the m -th smallest calibration score. In particular, $Q_\alpha(\emptyset) = +\infty$. Every rejection below uses a strict inequality against this threshold.

Lemma S1 (Exchangeable ranks). *Let $Z_1, \dots, Z_{n+1}$ be exchangeable real random variables. Then*

$$\Pr\{Z_{n+1} > Q_\alpha(Z_1, \dots, Z_n)\} \leq \alpha.$$

If $m \leq n$ and all $n+1$ scores are pairwise distinct almost surely, the probability equals $1 - m/(n+1)$ and lies in $(\alpha - 1/(n+1), \alpha]$. The same conclusions hold conditionally on a sigma-field relative to which the scores are conditionally exchangeable.

Proof. When $m = n+1$, the rejection event is empty. Otherwise put $N_i = |\{j \leq n+1 : Z_j < Z_i\}|$. The event $Z_{n+1} > Z_{(m)}$ is equivalent to $N_{n+1} \geq m$. Sort all $n+1$ scores, breaking ties arbitrarily. A score at position q has $N_i \leq q-1$, so at most $n+1-m$ indices satisfy $N_i \geq m$. Exchangeability therefore gives

$$\Pr(N_{n+1} \geq m) = \frac{1}{n+1} \mathbb{E}|\{i : N_i \geq m\}| \leq \frac{n+1-m}{n+1} \leq \alpha.$$

Without ties, exactly $n+1-m$ indices satisfy the inequality. The ceiling definition of m gives the asserted lower bound. Applying the same argument to the conditional law proves the conditional version. $\square$

Lemma S2 (Selection by a fixed criterion). *Let $W_1, \dots, W_{N+1}$ be exchangeable random elements, let $\mathcal{S}$ be a fixed measurable set, and put $J = \{i : W_i \in \mathcal{S}\}$. Conditional on any event $\{J = J_0\}$ of positive probability, the selected elements $(W_i)_{i \in J_0}$ are exchangeable. Thus, for a fixed measurable real score g , the upper conformal threshold of the selected calibration scores satisfies*

$$\Pr\{g(W_{N+1}) > Q_\alpha((g(W_i))_{i \in J \setminus \{N+1\}}) \mid N+1 \in J\} \leq \alpha$$

whenever the conditioning event has positive probability. The empty selected calibration set uses the threshold $+\infty$. These claims also hold conditionally on upstream information, provided exchangeability holds conditionally and $\mathcal{S}, g$ are fixed given that information.

Proof. Every permutation of the indices in J_0 that fixes its complement leaves the event $\{J = J_0\}$ invariant. Exchangeability before conditioning therefore implies exchangeability within J_0 after conditioning. Apply Lemma S1 to the selected scores for each J_0 containing $N+1$, with $n = |J_0| - 1$, and average the resulting bounds. This includes $n = 0$. The identical argument under the conditional law proves the last assertion. $\square$

Calibration protocol. Let $\mathcal{F}$ contain all training and tuning information. The addition-score function is trained on a separate subset of training runs not used to fit the scorer, response detectors or footprint operators, and on compositions of that subset. All these training subsets belong to $\mathcal{F}$. Given $\mathcal{F}$, the knowledge, footprint operators, scorer, encoder, response detectors, addition-score function, template means and covariance, tie-breaking rules, and fusion weight λ are fixed before any threshold-calibration pool is inspected. Use three further calibration pools C_0, C_1, C_u , with no run shared between pools or with training, tuning or evaluation. The pools are independent of one another conditional on $\mathcal{F}$. In each run choose one window using a rule independent of the run's content, before applying any correctness or score-based selection. The corresponding test window uses the same window-selection rule.

The required distributional assumptions are conditional exchangeability at each stage. Given $\mathcal{F}$, the fault-free windows in C_0 and a fault-free test window are exchangeable. Given $\mathcal{F}, C_0$, the labeled single-event windows in C_1 and a single-event test window are exchangeable. Given $\mathcal{F}, C_0, C_1$, the windows in C_u and a test window from its specified mixture of normal operation and known single events are exchangeable. Independent identically distributed runs from each stated population, conditional on training and tuning information, are a sufficient way to obtain these conditions. Disjoint run identifiers alone do not imply them; in particular, a fixed allocation of class counts is not by itself an exchangeability argument for a pooled mixture.

First set $\tau_0 = 1 - Q_{\alpha_0}((1 - p_\emptyset(X_i))_{i \in C_0})$, interpreting $1 - (+\infty) = -\infty$, and freeze τ_0 . For a labeled single-event window $W = (X, E)$, define

$$A(W) = \mathbf{1}\{p_\emptyset(X) < \tau_0, \arg \max_{E' \in \mathcal{E}} p_{E'}(X) = E\}.$$

This criterion includes the normal-operation gate and is fixed given $\mathcal{F}, C_0$. On the windows in C_1 with $A(W_i) = 1$, compute

$$T^*(W_i) = \max_{E' \neq E_i} z_{E'}(X_i, \phi_i - \hat{\Delta}_{E_i}(\phi_i), \{E_i\}), \quad \tau_1 = Q_{\alpha_1}((T^*(W_i))_{i \in C_1: A(W_i)=1}).$$

Assume the library contains at least two events for this definition; with only one event no false addition is possible and take $\tau_1 = +\infty$. Freeze τ_1 and the resulting output rule $H(X)$. Finally set

$$\tau_u = Q_{\alpha_u}((D_{H(X)}(\phi(X)))_{i \in C_u}).$$

All scores used here are assumed finite and measurable. The bound $K_{\max}$ is at least one.

Theorem 4 (Finite-sample control of false naming). *For a fixed fusion weight λ , under the preceding protocol and its conditional exchangeability assumptions, the following bounds hold, averaging over the relevant calibration pool and test window. (a) On a fault-free test window, the probability that AEC-X names an event is at most α_0 . If the $n_0 + 1$ normal scores have no ties and $m_0 = \lceil (1 - \alpha_0)(n_0 + 1) \rceil \leq n_0$, this probability is $1 - m_0/(n_0 + 1)$ and is greater than $\alpha_0 - 1/(n_0 + 1)$. (b) Conditional on a single-event test window passing the normal-operation gate and selecting its true event first, the probability of adding any false event is at most α_1 , whenever this conditioning event has positive probability. (c) For a test window from the same specified mixture as C_u , the probability of an unknown flag is at most α_u . Ties are permitted for all three upper bounds.*

Proof. (a) Given $\mathcal{F}$, the n_0 normal calibration scores and the normal test score $Z = 1 - p_\emptyset(X)$ are exchangeable. Naming occurs exactly when $Z > Q_{\alpha_0}$, including the case $\tau_0 = -\infty$, in which it never occurs. Lemma S1 gives the conditional upper bound, and its equality statement gives the claimed formula when ties are absent and $m_0 \leq n_0$. Averaging over $\mathcal{F}$ gives the marginal statements.

(b) Given $\mathcal{F}, C_0$, both A and T^* are fixed functions of a labeled single-event window. Lemma S2 therefore gives, for almost every upstream realization on which the test selection event has positive probability,

$$\Pr\{T^*(W_{\text{test}}) > \tau_1 \mid \mathcal{F}, C_0, A(W_{\text{test}}) = 1\} \leq \alpha_1.$$

If the first selected event is the true one and a second selection is allowed, every remaining candidate is false, so a false second event is added exactly when $T^* > \tau_1$. Otherwise the procedure stops immediately with the true singleton and cannot add a false event at a later step. When $K_{\max} = 1$, or the library contains only one event, the false-addition event is empty. Hence the desired event is always contained in the score-exceedance event in the nontrivial case. Averaging the conditional bound over upstream information given $A(W_{\text{test}}) = 1$ proves (b).

(c) Given $\mathcal{F}, C_0, C_1$, the entire output rule H , including both selection thresholds, is fixed before C_u is inspected. Thus $X \mapsto D_{H(X)}(\phi(X))$ is a fixed measurable score under this conditioning. The assumed conditional exchangeability of the known-mixture calibration and test windows makes their scores exchangeable. The flag uses $D_{H(X)}(\phi(X)) > \tau_u$, so Lemma S1 gives the conditional upper bound; averaging gives (c). $\square$

Remark S7 (Scope, fluctuation and implementation). Part (c) controls the stated mixture marginally, not each output set or each event class. Conditional control within an output group would require an appropriate groupwise calibration construction and sufficient selected calibration data. Known concurrent-event windows have no guarantee from (c) unless their distribution also satisfies its exchangeability hypothesis. Their false-flag rate must otherwise be evaluated empirically.

For the Beta fluctuation statement, impose the stronger assumption that, conditional on $\mathcal{F}$, the normal calibration and test scores are independent and identically distributed with continuous distribution function $F_{\mathcal{F}}$, and $m_0 \leq n_0$. The conditional test naming probability

$$q(C_0) = \Pr(\text{naming} \mid \mathcal{F}, C_0) = 1 - F_{\mathcal{F}}(Z_{(m_0)})$$

then has distribution $\text{Beta}(n_0 + 1 - m_0, m_0)$ over random draws of C_0 , conditional on $\mathcal{F}$: the transformed calibration scores are independent uniform variables, whose m_0 -th order statistic is $\text{Beta}(m_0, n_0 + 1 - m_0)$ [5,6]. Once C_0 is fixed, $q(C_0)$ is a number. Mere exchangeability does not imply this Beta law. When $m_0 = n_0 + 1$, $q(C_0) = 0$ under the infinite-threshold convention.

The protocol above is sufficient for this theorem, not a claim that every released experiment follows it. Selecting the fusion weight on a threshold-calibration pool, or reusing the pools that determine H to calibrate the unknown score, is not justified by this proof. The repository’s proof README records the implementation scope; disjointness from evaluation data alone does not supply the missing conditional exchangeability.

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